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CS604 SOLVED MCQS FINAL TERM

Date 10th Feb 2013

_____ commands in Linux is used to copy file
Is
Cp (p27)
mv
mkdir

The correct command for compiling C program named program.c in Linux environment is
gcc program.c –o FirstPrgram (p28)
gcc –o FirstProgram program.c
gcc –z FirstProgram program.c
gcc program.c –m FirstPrgram
Bottom of Form
Top of Form

Question # 1
_____ scheduler selects the process from the job pool and put them in main memory.
Select correct option:
Long term (p33)
Short term
Medium term
Swapper

Question # 2
The process id returned to the child process after successful fork system call execution is _____.
Select correct option:
0 (p36)
1
2
3

Question # 6
_____ displays information about the top processes.
Select correct option:
ls
cs
top (p64)
cd

Question # 8
Using _____system, we can create a new process in Linux.
Select correct option:
fork (p36)
exec

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wait
exit

Question # 3
_____ commands in Linux helps to create a new directory.
Select correct option:
ls
cp
mv
mkdir (p26)

Cooperating processes never share any data, code, memory or state.
Select correct option:
True
False (p41)

Question # 7
The manual pages can be read in Linux using ____ command.
Select correct option:
man (p4)
wan
desc
help

Question # 8
In Unix/ Linux, by default the standard input file is attached to the _____
Select correct option:
Mouse
Keyboard (p52)
Light pen
Joystick

Standard input: 0 (keyboard)
Standard output: 1 (display screen)
Standard error: 2 (display screen)

Question # 2
Shared libraries and kernel modules are stored in _____ directory.
Select correct option:
/bin
/dev
/boot
/lib (p23)

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Question # 7
Swapper is also termed as Short term scheduler.
Select correct option:
True
False (p34) (Medium term = Swapper)

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Question # 10
You can use the mv file1 file2 command to move _____
Select correct option:
file1 to file2. (p27)
file 2 to file 1
this command will not work for moving files
None of the option is correct.

Question # 2
_____ is also called Swapper.
Select correct option:
Long term
Short term
Medium term (p34)
Swap space

Question # 3
_____ system call is used to write to a file or FIFO or any other IPC channel.
Select correct option:
read
write (p45)
open
fork

Question # 6
If your processor does not have two slots empty in Per Process File Descriptor Table, then your _____ system call will fail.
Select correct option:
Pipe (p52)
Read
Write
Open

The pipe() system call fails for many reasons, including the following:
-At least two slots are not empty in the PPFDT—too many files or pipes are open in the process
-Buffer space not available in the kernel
-File table is full

2. The Operating system is a layer of software between _____ and _____.
Hardware, software application
Kernel, hardware
Dos, Windows
Windows, Kernel

5. Command-line interpreter is also called _____ in some operating systems.
a) Kernel
b) Shell (p13)
c) Signal
d) API

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6. I/O instructions are Privileged Instructions.
a) True (p8)
b) False

7. In Linux directory structure, there is _____ root directory.
a) 1 (p22)
b) 2
c) 3
d) 4

8. Utilities used for system administration (halt, ifconfig, fdisk, etc.) are stored in _____ directory.
a) /dev
b) /boot
c) /lib
d) /sbin (p24)

9. rm and rmkdir commands are used to _____ directory.
a) Create
b) Move
c) Remove (p27)
d) Modify

15. In _____ communication the process which wants to communicate with the other process must explicitly name the recipient and the sender.
a) Direct (p43)
b) Indirect
c) Automatic
d) Self

16. In indirect inter process communication, a sender _____ mention the name of the recipient.
a) does
b) does not (p43)

17. The returned code to the child process after successful fork system call execution is
a) 1
b) 2
c) 3
d) 0 (p32)

18. If the fork system call fails, it returns
a) 1
b) -1 (p32)
c) 2

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d) 0

Question # 1
The problem with priority scheduling algorithm is _____.
Select correct option:
Deadlock
Starvation (p84)
Aging
Nice value

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- A major problem with priority- scheduling algorithms is indefinite blocking (or starvation).
- Aging is solution to the problem of indefinite blockage of low-priority processes.

A major problem with priority scheduling algorithms is _____.
Select correct option:
Deadlock
Aging
Starvation (P # 84)
None of these

_____ is a solution to the problem of indefinite blockage of low-priority processes.
Starvation
Deadlock
Aging (p85)
None of the these

Question # 4
Batch programs are usually _____ programs.
Select correct option:
Interactive
Non-interactive (Google)
Foreground
Preemptive

Question # 2
All Threads within a process share the _____ address space.
Select correct option:
Same (p70)
Different

Question # 6
_____ scheduling allows a process to move between queues.
Select correct option:
Round Robin
First Come First Serve
Multilevel Feedback Queue (p90)

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Shortest Remaining Time First

Question # 4
POSIX is a standard developed by _____
Select correct option:
ANSI
IEEE (Reference)
ISO
ACM

Question # 5
The nice value helps in assigning _____ to a process.
Select correct option:
Priority (Google)
Weight
Time
Scheduling

Question # 9
_____ integer shows the highest priority of a process in CPU scheduling
Select correct option:
Small (p84)
Large

Question # 3
____ command display the status of a process.
Select correct option:
ls
ps (p63)
gcc
cat

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In Unix/ Linux, by default the standard output file is attached to the _____
Select correct option:
File
Screen (p52)
Printer
Scanner

The priority of a process can be changed using _____ command.
Select correct option:
Nice (p # 92)
cmd
cat
grep

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Question # 8
_____ is a preemptive scheduling algorithm.
Select correct option:
First Come First Serve
Shortest Job First
Round Robin (p87)
None of these

- FCFS is a non-preemptive scheduling algorithm.
- The SJF algorithm may either be preemptive or non-preemptive.
- Preemptive SJF scheduling is sometimes called shortest-remaining-time-first scheduling.
- SJF is a special case of the general priority-scheduling algorithm.
- Priority scheduling can either be preemptive or non-preemptive.
- Round Robin is preemptive scheduling algorithm.
- Round-robin scheduling algorithm is designed especially for time-sharing systems.
- RR is similar to FCFS scheduling but preemption is added to switch between processes.

Question # 8
_____ is used in real time operating systems.
Select correct option:
Non-preemptive scheduling
Preemptive scheduling (Reference)
Dispatching scheduling
FCFS scheduling

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Question # 10
The procedure "The time at which the process finished working MINUS the arrival time of the process MINUS CPU burst for that process" will help calculate the _____.
Select correct option:
Non-preemptive Shortest Job First scheduling.
Preemptive Shortest Job First scheduling. (p82,83)
FCFS
RR Scheduling

Non-Preemptive => Waiting time = Finish Time – Arrival Time
Preemptive => Waiting time = Finish Time – Arrival Time – Burst time

Question # 9
_____ scheduling algorithm is sometimes called shortest remaining time first scheduling algorithm.
Select correct option:
Non-preemptive SJF
Priority Scheduling
Preemptive Shortest Job First (P #83)
FCFS

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_____ scheduling algorithm can be preemptive or non-preemptive.
First Come First Serve
Shortest Job First (P # 83)
Round Robin
Priority

Question # 7
Round Robin algorithm is similar to _____ scheduling but preemption is added to switch between processes.
Shortest jab first
Shortest remaining time first
First come first serve (P # 86)
None of theses

Question # 10
Round Robin algorithm is most suitable for _____.
Time sharing system (p86)
Real time systems and batch systems
Running batch programs
Expert system

Question # 1
Kernel is responsible for scheduling the user level threads.
True
False (Reference)

The kernel is responsible for scheduling the kernel threads, so it effectively schedules the user threads at the same time.

Question # 2
The priorities of processes in the _____ group remain fixed.
Select correct option:
Kernel (p # 91)
User

Question # 4
_____ is the basis of queuing theory which is branch of mathematics used to analyze systems involving queues and servers.
Little's formula (p # 94)
Deterministic theory
Queuing theory
Queuing analysis

_____systems involving queues and servers.
Little's formula (p # 94)
Deterministic Modeling

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Queuing Theory
Queuing analysis

Question # 9
The scheduling of _____ are done by the operating system.
Kernel threads (p71)
User level threads
Both
None of these
- Kernel threads are supported directly by the operating system.

Question # 10
Taking the CPU from one process and giving the CPU to another process is termed as
Select correct option:
Context switching (p87)
Dispatching
Swapping
Tracking

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Question # 10
The process of switching from one process to another is called latency
True
False

The process of switching from one process to another is called ----------
Context switching (p87)
Scheduling
Quantum period
Latency

The time it takes for the dispatcher to stop one process and start another running is known as the **dispatch latency**.
The **rotational latency** is the additional time waiting for the disk to rotate the desired sector to the disk head.
I/O time is composed of seek, latency and transfer times. Transfer time is proportional to the amount transferred, and this argues for a small page size. However, latency and seek times usually dwarf transfer times.

Question # 12
The time it takes for the dispatcher to stop one process and start another running is known as the----------.
Dispatch latency (p80)
Scheduling
Context switching
None of the given options

Question No:

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The hardware mechanism that enables a device to notify CPU is called an ----------
Interrupt (p553 Operating Systems Concepts)
Signal
Trap
Process

Question # 1
Address Binding will be at _____ in Multiprogramming with Fixed Tasks (MFT)
Select correct option:
Run time
Load time (p161)
Dynamic time
None of the given options

Question # 2
In ---------- technique, memory is divided into several fixed-size partitions.
Select correct option:
Swapping
Overlays
Multiprogramming with Fixed Tasks (MFT) (p160)
Multiprogramming with Variable Tasks (MVT)

Question # 4
----------holds the smallest legal physical memory address for a process
Select correct option:
Base register (p10)
Limit register
Index register
Stack pointers register

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Question # 5
----------refers to the situation when free memory space exists to load a process in the memory but the space is not contiguous.
Select correct option:
Segmentation
Internal fragmentation
Swapping
External Fragmentation (p163)

Question # 7
Main Memory is _____ memory.
Select correct option:
-Volatile
Non-volatile
Permanent
Virtual

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Overlays:
To enable a process to be larger than the amount of memory allocated to it, we can use overlays. The idea of overlays is to keep in memory only those instructions and data that are needed at any given time. When other instructions are needed, they are loaded into space occupied previously by instructions that are no longer needed.

Overlays are implemented by the _____
Operating system
Programmer (p157)
Kernel
Shell

Question # 13
----------keep in memory only those instructions and data that are needed at any given time.
Select correct option:
Fragmentation
Paging
Swapping
Overlays (p157)

Question # 10
Secondary Storage memory devices have _____ memory.
Select correct option:
Volatile
Permanent and non-volatile (p151)
Temporary
None of the options is correct

Question # 12
_____ algorithm is used in Deadlock avoidance.
Select correct option:
Bakery
Banker's (p137)
Mutual exclusion
Safe Sequence

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Question # 14
Addresses generated relative to part of program, not to start of physical memory are
Select correct option:
Relocatable (page334 Operation systems concepts)
Virtual
Symbolic
Physical

Question # 15

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The run-time mapping from virtual to physical addresses is done by a piece of hardware in the CPU, called the ----------
Select correct option:
Memory management unit (MMU) (p152)
CPU Scheduler
Registers
None of the given options

Question # 2
_____ algorithm is used for solving n-process critical section problem.
Select correct option:
Bankers
Bakery (p103)
Babbles
None of the given

Question # 4
_____ is an integer variable accessible through wait and signal which are atomic operations.
Select correct option:
Semaphore (p109)
Mutex
Busy waiting
Signal

Question No: 2
You can display the contents (names of files and directories) of a directory in UNIX/Linux directory structure with the ---------- command.
ll
s
ls (p25)
None of the given options

Question No: 3
The ---------- system call suspends the calling process.
fork
wait (p39)
exec
exit

Question No: 4
In ---------addressing, the recipient is not required to name the sender.
Symmetric
Asymmetric (p44)
Both symmetric and asymmetric
None of the given options

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Question No: 10
The main characteristic of a Real time system is
Efficiency (p7)
Large Virtual Memory
Large secondary storage device
Usability

Question No: 14
A ____is an integer variable that, apart from initialization is accessible only through two standard atomic operations: wait and signal.
Semaphore (p109)
Monitor
Critical region
Critical section

Question No: 1
1 MB or 1 megabyte is equivalent to----**1048576 bytes**
1024 bytes
1024^2 bytes
1024^3 bytes
1000000 bytes

Question No: 2
---------- spend more time doing IO than computations short CPU bursts
CPU bound processes
IO bound processes (p29)
None of the given options

With ----------you use condition variables.
Semaphores
Read/Write Locks
Swaps
Monitor (p124)

Deadlocks can be described more precisely in terms of a directed graph called a system ----------
Directed graph
Critical path
Resource allocation graph (p130)
Mixed graph

Object files and libraries are combined by a ---------- program to produce the executable binary
Compiler
Linker (p428,818 Operating system concepts)
Text editor

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Loader

The set of all physical addresses corresponding to the logical addresses is a ---------- of the process
Physical address space (p153)
Process address space
None of the given options
Logical address space

---------- indicates size of the page table.
translation look-aside buffers
Page-table length register (PTLR) (p350 Operating system concepts)
Page-table base register (PTBR)
Page offset

Question
----------points to the page table.
Translation look-aside buffers
Page offset
Page-table length register (PRLR)
Page-table base register (PTBR) (p347 OSC)

If validation bit is 0, it indicates a/an --------- state of segment.
Protected
Shared
Legal
Illegal (p178)

Question No: 11
In _____ allocation scheme free frames are equally divided among processes
Fixed Allocation (p205)
Proportional Allocation
Priority Allocation
None of the given options

_____ is used to store data on secondary storage device, e.g., a source program(in C), an executable program.
Block Special File
Link File
Ordinary File (p218)
Directory

The _____ method requires each file to occupy a set of contiguous blocks on the disk.
Contiguous Allocation (p234)
Linked Allocation

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Indexed Allocation
None of the given options

User mode can run the Privileged instructions
True
False (p8)

Privileged instructions can be used to perform hardware operations that a normal user process should not be able to perform, such as communicating with I/O devices. If a user process tries to execute a privileged instruction, a trap should be generated and process should be terminated prematurely.

_____ wastes CPU cycles and hence is a problem in real multiprogramming system.
Busy waiting (p256 OSC)
Spinlock
Critical section
Mutex

This continual looping is clearly a problem in a real multiprogramming system, where a single CPU is shared among many processes. Busy waiting wastes CPU cycles that some other process might be able to use productively. This type of semaphore is also called a spinlock because the process "spins" while waiting for the lock.

The _____ requires that no reader will be kept waiting unless a writer has already obtained permission to use the shared object.
first readers-writers problem (p116)
second readers-writers problem
third readers-writers problem
fourth readers-writers problem

The first readers-writers problem, requires that no reader will be kept waiting unless a writer has already obtained permission to use the shared object. In other words, no reader should wait for other readers to finish simply because a writer is waiting.
The second readers-writers problem requires that once a writer is ready, that writer performs its write as soon as possible. In other words, if a writer is waiting to access the object, no new readers may start reading.

The process of holding at least one resource and waiting to acquire additional resources that are currently being held by other processes is known as_____.
Mutual exclusion
Hold and wait (p129)
No preemption
Circular wait

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If a process continues to fault, replacing pages, for which it then faults and brings back in right away. This high paging activity is called _____.
Paging

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Thrashing (p208)
Page fault
CPU utilization

Question # 7
The high paging activity is called _____
Select correct option:
Segmentation
Page Fault
Multiprogramming
Thrashing (p208)

A process is _____if it is spending more time on paging
Select correct option:
Thrashing
Demand paging
Paging
Fixed Allocation

Banker's algorithm is used for _____
Deadlock avoidance (p313 OSC)
Deadlock detection
Deadlock prevention
Deadlock removal

A program cannot execute unless whole or necessary part of it resides in the main memory.
True (not sure)
False

The size of pages and frames are same in logical memory and physical memory respectively.
True
False

Which command, Display permissions and some other attributes for prog1.c in your current directory?
ls –l prog1.c
ls –d prog1.c
ls file prog1.c
ls –l prog1.c /Directory

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In the C-Scan and C-Look algorithms, when the disk head reverses its direction, it moves all the way to the other end, without serving any requests, and then reverses again and starts serving requests.
True
False

In paged segmentation, we divide every segment in a process into _____pages.
Fixed size (p180)

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Variable size

Intel 80386 used paged segmentation with _____ level paging.
One
Two (p183)
Three
Four

The logical address of Intel 80386 is _____
36 bits
48 bits (p183)
64 bits
128 bits

Following is NOT true about Virtual memory.
Virtual memory help in executing bigger programs even greater in size that of main memory.
Virtual memory makes the processes to stuck when the collective size of all the processes becomes greater than the size of main memory.
Virtual memory also allows files and memory to be shared by several different processes through page sharing.
Virtual memory makes the task of programming easier because the programmer need not worry about the amount of physical memory,

The Swap instruction which is the hardware solution to synchronization problem does not satisfy the _____ condition, hence not considered to be a good solution.
Progress
Bounded waiting
Mutual exclusion
None of the given

Summarize the tradeoffs among simple arrays, trees, and hash tables as implementations of a page table.
A ---------- (or an *exception)* is a software-generated interrupt caused either by an error (division by zero or invalid memory access) or by a user request for an operating system service.
Interrupt
Trap
Signal
Process

Which register holds the smallest legal physical memory address for a process?
Base register
Limit register
Status register
None of the given options

The ----------semaphore provides mutual exclusion for accesses to the buffer pool and is initialized to the value 1.

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mutex
Binary
Couting
None of the given options

Binary semaphores are those that have only two values---------
0 and n
0 and 0
0 and 1
None of the given options

Physical memory is broken down into fixed-sized blocks, called---------- and
Logical memory is divided into blocks of the same size, called ----------
Frames, pages
Pages, Frames
Frames, holes
Holes, segments

A page table needed for keeping track of pages of the page table is called
----
2-level paging
Page directory
Page size
Page table size

The address generated by the CPU, after any indexing or other addressing-mode arithmetic, is called a ----------*address*, and the address it gets translated to by the
MMU is called a ---------*address*.
Virtual, physical
Hexadecimal, Binary,
Valid, invalid
Physical, Virtual

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Question No: 11
Each page is a power of -------- bytes long in paging scheme.
2 (p163)
3
4
5

Question No: 12
_____ is a way to establish a connection between the file to be shared and the directory entries of the users who want to have aces to this file.
Link (p229)
Directory
Common Group

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Access Permission

When a _____link is created, a directory entry for the existing file is created
Soft
None of the given options
Hard (p225)
Soft or Hard
Soft or Hard

Which part of the computer system helps in managing the file and memory management system?
Operating System (p242)
Device Drivers
Application Software
Hardware

Wrong use of wait and signal operations (in context with semaphores) can cause____ problem(s).
Mutual Exclusion
Deadlock
Bounded Waiting
All of the given options (p114)

If a system is not in a safe state, there can be no deadlocks.
True
False

In _____ page replace algorithm we will replace the page that has not been used for the longest period of time.
Counter based
Least Frequently Used
FIFO
LRU (p200)

An acyclic graph does not allow directories to have shared subdirectories and files.
True
False (p223)

A modification of free-list approach in free space management is to store the addresses of n free blocks in the first free block. Known as _____.
Counting
Linked list
Bit vector
Grouping (p241)

In deadlock detection and recovery algorithm, a deadlock exists in the system if and only if the wait for graph contains a _____
Cycle

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Graph
Edge
Node

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Intel is basically designed for following Operating Systems except _____.
MULTICS
OS/2
Windows
Linux

The bottom layer in the layered approach of Operating System is----------
User interface
Hardware
Kernel
None of the given options

In which of the following operations, the scheduler is not called into play?
Process requests for I/O.
Process finishes execution.
Process finishes its time allotted.
All of the above through c
None of the a through c above.

Consider the following preemptive priority-scheduling algorithm based on dynamically changing priorities. Larger priority numbers imply higher priority. When a process is waiting for the CPU (in the ready queue but not running), its priority changes at a rate X when it is running, its priority changes at a rate Y. All processes are given a priority of 0 when they enter the ready queue. The parameters and can be set to give many different scheduling algorithms. What is the algorithm that results from Y>X>0?
LIFO
FCFS
Round Robin
None of the above

A CPU scheduling algorithm determines an order for the execution of its scheduled processes. Given *n* processes to be scheduled on one processor, how many possible different schedules are there? Give a formula in terms of *n*.
n(n-1)
n2
n!
n/2

"To eliminate external fragmentation in segmentation the scheme used is"

Segmentation is a memory management scheme that supports _____?
Programmer's view of memory
System's view of memory

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Hardware's view of memory
None of the given

---------- register contains the size of the process
Base register
Index register
Limit register
Stack pointers register

Preventing the condition of _____ to happen, deadlocks can be prevented to happen
Critical Region
Circular wait
Monitors
Critical Section

Assume a logical address space of 16 pages of 1024 words, each mapped into a physical memory of 32 frames. Each word consists of 2 bytes. What will be the total number of bits required for p (page number)?
4 bits
8 bits
16 bits
32 bits

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A dashed line is used to represent a _____ in Resource Allocation Graph
Claim edge
Request edge
Assignment edge
Allocation edge

Question # 1
The main memory is usually divided into partitions, one for _____ and other for _____ .
Select correct option:
Operating System, User processes
Operating system, CPU
Processes, Virtual Memory
Base Register, Limit Register

_____ is used in the detection and recovery mechanism to handle deadlocks. Select correct option:
Wait-for Graph
Resource allocation Graph
Circular Graph
Claim Edge Graph

_____ is caused due to un-used space in fixed size blocks/ pages. Select correct option:

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Internal fragmentation
External fragmentation
Paging
MVT

In Swapping technique of Memory Management, the total amount transfer time is directly proportional to the _____.
Select correct option:
Amount of memory swapped
Amount of space on backing store
space on main memory
All the given options are correct

What do we name to an address that is generated by the CPU? Select correct option:
Logical address
Physical address
Binary addresses
None of the given options

The ----------scheme is not applicable to a resource allocation system with multiple instances of each resource type.
Select correct option:
Wait for graph
Resource allocation graph
Both Resource-allocation and wait-for graph
None of the given options

Memory protection in paging is achieved by associating _____ with each page. This bit indicates whether the page is in the process address space or not.
Select correct option:
Protection bits
Counting bits
Paging table
segment table

In Resource Allocation Graph**,** A _____ Pi →Rj indicates that process Pi may request resource Rj at some time in the future.
Select correct option:
Claim edge
Request edge
Assignment edge
Allocation edge

_____ is caused due to un-used space in physical memory
Internal fragmentation
External fragmentation

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Paging
MVT

Every ----------generated by the CPU is divided into two parts: a page number (p) and a page offset (d).
Select correct option:
Page
Process address space
Physical address
Logical address

What do we name to an address that is loaded into the memory-address register of the memory?
Select correct option:
Logical address
Physical address
Binary addresses
None of the given options

Variable names are ---------- addresses
Select correct option:
Physical
Relocatable
Relative
Symbolic

The collection of processes that is waiting on the disk to be brought into the memory for execution forms the --------
Select correct option:
Input queue
Output queue
Both input and output queue
None of the given options

In Swapping technique of Memory Management, the total amount transfer time is directly proportional to the _____.
Select correct option:
Amount of memory swapped
Amount of space on backing store
Space on main memory
All the given options are correct

Question # 5
What do we name to an address that is generated by the CPU?
Select correct option:
Logical address
Physical address

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Binary addresses
None of the given options

The run-time mapping from virtual to physical addresses is done by a piece of hardware in the CPU, called the ----------
Select correct option:
Memory management unit (MMU)
CPU Scheduler
Registers
None of the given options

What do we name to an address that is loaded into the memory-address register of the memory?
Select correct option:
Logical address
Physical address
Binary addresses
None of the given options

If the system can allocate resources to each process in some order and still avoid a deadlock then it is said to be in _____ state
Safe
Un-Safe
Mutual
Starvation

_____is a process for mapping a name to an address.
Addressing
Binding
Routing
Memory

When the address used in a program gets converted to an actual physical RAM address, it is called ----------
Select correct option:
Execution
Loading
Address Binding
Compiling

----------holds the smallest legal physical memory address for a process
Select correct option:
Base register (p331)
Limit register
Index register
Stack pointers register

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Main Memory is _____ memory.
Select correct option:
Volatile
Non-volatile
Permanent
Virtual

Question # 8
In_____, the library files are linked at load time.
Select correct option:
Static Linking
Dynamic Linking

The ----- is a single program that produces an object file
Select correct option:
Linker
Compiler (Ref)
Loader
Text editor

Cache is non-volatile memory.
Select correct option:
True
False

LRU page replacement algorithm can be implemented by
Select correct option:
Counter
Stack
Linked list
All of the given options (p201)

Question # 3
Assume a logical address space of 16 pages of 1024 words, each mapped into a physical memory of 32 frames. Each word consists of 2 bytes. What will be the total number of bits required for p (page number)?
4 bits (p166) No. of bits needed for p = ceiling [log2 16] bits = 4 bits
8 bits
16 bits
32 bits

Question # 4
Assume a logical address space of 16 pages, each of 1024 words, each mapped into a physical memory of 32 frames. Each word consists of 2 bytes. What will be the total number of bits required for f (frames)?
Select correct option:
5 (p166) No. of bits neede for f = ceiling [log2 32] bits = 5 bits

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6
7
8

Question # 5
In _____ page replacement algorithm we will replace the page that has not been used for the longest period of time.
Select correct option:
Counter based
Least Frequently Used
FIFO
LRU (p200)

Question # 6
_____is the separation of user logical memory from physical memory.
Select correct option:
ROM
Physical memory
Virtual Memory (p186)
None of the given options

Question # 8
Following are the advantages of Virtual memory EXCEPT
Select correct option:
Efficient process creation
Concept of memory mapped files
Low CPU consumption (p186)
Running large sized processes

Question # 9
A _____ system is similar to a paging system with swapping
Select correct option:
Context switching
Demand paging (p187)
Page fault
None of the given options

Question # 10
In _____ allocation scheme number of frames allocated to a process is proportional to its size .
Fixed Allocation
Proportional Allocation (p204)
Priority Allocation
None of the given options

Question # 2
The major criterion in the selection of a particular algorithm is that we want to

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Select correct option:
Minimize the number of page faults. (p193)
Increase efficiency.
Reduce running time of page replacement algorithm.
Maximize the number of page faults.

_____is the operating system's attempt to improve the computer system's Utilization and throughput.
Select correct option:
Exec
Fork
Demand Paging (p197)
Thrashing

Question # 7
To eliminate external fragmentation in segmentation the scheme used is _____.
Select correct option:
Fixed size partition scheme
Variable size partition scheme
Fixed size paging scheme
Variable size paging scheme

Question # 9
In a UNIX system,_____ system call can be used to request the operating system to memory map an opened file.
Select correct option:
mmap( ) (p196)
fork ( )
exec( )
read( )

Question # 2
For some page replacement algorithms, the page fault rate may increase as the number of allocated frames _____.
Select correct option:
Constant
Decreased
Increased (p198)
All of the given

_____is a variation of fork system call in several Unix operating system.
Select correct option:
vfork ( ) (p194)
wfork ( )
avfork ( )
bfork ( )

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The main criteria for Page Replacement in Optimal Page Replacement Algorithm is to
Select correct option:
Replace that page which will not be used for the longest period of time. (p200)
Replace that page which will be required most frequently in the execution of a process
Replace the page which is biggest in size
Replace the page which is smaller in size

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Question # 7
In pages segmentation, the logical address is legal if d is _____segment length.
Select correct option:
Less than (p176)
Greater than
Equal to
Greater than or equal to

- Segment number s is legal if s < STLR, and offset, d, is legal if d < limit or Segment Length.

The frame that has been modified during execution of a process is usually called _____ frame.
Select correct option:
Clean
Dirty (Reference)
Overwritten
Ideal

FIFO Page Replacement Algorithm has the lowest page fault rate
Select correct option:
True
False (p198, 200)

- An optimal page-replacement algorithm has the lowest page fault rate.

When the process tries to access locations that are not in memory, the hardware traps the operating system. This is called as _____.
Select correct option:
Page Fault (p190)
Page replacement
Paging
Segmentation

Following schemes allow efficient implementations of page tables EXCEPT
Select correct option:
Hashed Page Table
Hierarchical / Multilevel Paging
Inverted Page Table
Binary Page Table (p171)

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Question # 1
We want a page replacement algorithm with the _____ page-fault rate.
Select correct option:
Highest
Lowest (p199)
Normal
None of the given

Question # 10
In case of thrashing if CPU utilization is too low the operating system _____ the degree of multiprogramming.
Select correct option:
Increases (p208)
Decreases
Sometimes increases and sometimes decreases
None of the given options

It is **not possible** to run a program whose size is greater than the size of the main memory.
Select correct option:
True
False (Possible with Virtual Memory Concept)

Question # 6
An optimal page-replacement algorithm has the lowest page fault rate of all algorithms.
Select correct option:
True (p200)
False

A solution to the critical section problem must satisfy the following requirements except:
Progress
Mutual Exclusion
Bounded Waiting
Race Condition (p99)

The solution to Critical Section Problem is : Mutual Exclusion, Progress and Bounded Waiting.

Question # 10
A process is said to be in critical section if it executes code that manipulates shared data.
True
False

Question No: 7
Logical address is generated by:
CPU (Reference)

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Compiler
Hard disk
None of the these

Consider a UNIX system with **threshold priority** of 125. Assume two processes, P1 and P2, which came into the system at the same time. P1 has a **nice value** of 15 and 'recent CPU usage' 45 ticks. P2 has a nice value of 10 and 'recent CPU usage' 65 ticks. It is time for scheduling. Which of the two processes will be chosen for execution?
P1 because its recent CPU usage is less than that of P2's and, therefore, its priority is higher than P2's priority.
P2 because its nice value is smaller than that of P1's. (Reference)
P2 because its priority number is smaller than that of P1's.
P1 because its process ID is smaller than P2's process ID.
None of the given choices
.
Question No: 2
What is spooling?
Ability of an OS to protect jobs from writing into the wrong memory location
Ability of an OS to do long term job scheduling
Ability of an OS to read jobs from cards onto the disk, and load a new job from the disk to empty memory partition
Ability of an OS to give priority to each job for execution.
None of the given choices.

- Spooling is the process of writing the contents of a print job to a file on disk. This file is called a *spool file*.
- When the job requests the printer to output a line, that line is copied into a system buffer and is written to the disk. When the job is completed, the output is actually printed. This form of processing is called spooling.
- Spooling overlaps the I/O of one job with the computation of other jobs.
- Spooling is also used for processing data at remote sites.
- Spooling has a direct beneficial effect on the performance of the system.

Question No: 3
To a computer operating system, a thread is
Trace of system calls made by a process.
Identifiable computation unit with state executing a task.
Input/output stream associated with a process
All of the given choices.
None of the given choices.

- A thread is a basic unit of CPU utilization; it comprises a thread ID, a program counter, a register set, and a stack.
- A thread is a flow of control within a process.
- A thread is a unit of code that can be scheduled by the operating system.

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Question No: 4
Which component ensures that a process can execute only within its own address space?
I/O device
Memory-addressing hardware (p80 OSC)
Timer
Virtual memory
None of the given choices.

Segmented paging incurs less internal fragmentation than pure process-level paging.
True
False (Reference)
Ambiguous question

Question No: 2
The Multi-Level Feedback Queue (MLFQ) scheduling algorithm is the same as Shortest-Job-First.
True
False (Reference)
Ambiguous question

a) In paging systems, external fragmentation cannot occur. **TRUE**

b) Race conditions cannot occur on a uniprocessor. **FALSE**

c) SJF can be implemented as a priority algorithm, where the priority is determined by the arrival time of the job. **FALSE**

d) A process in the Ready state can only transition to Running or Exit states. **TRUE**

e) The two-phase locking protocol guarantees that concurrent transactions are deadlock-free. **FALSE**

(References for a,b,c,d,e above)

Question # 6
A condition where a set of blocked processes each holding a resource and waiting to acquire a resource held by another process in the set is termed as _____.
Select correct option:
Deadlock (p128)
Starvation

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Question # 7
Preventing a condition of _____ to happen, deadlocks can be prevented to happen.
Select correct option:
Critical region
Circular wait (p134)
Monitors

Deadlock Prevention
1. Mutual exclusion
2. Hold and Wait
3. No preemption
4. Circular Wait

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Critical section

Question # 8
The following is NOT a classical problem of synchronization
Select correct option:
Bounded buffer problem
Reader writer problem
Dining philosopher's problem
Counting semaphore problem (p116)

Question # 9
The integer value of _____semaphores can range over an unrestricted integer domain.
Select correct option:
Counting (p115)
Binary
Mutex
Bounded buffer

There are two kinds of semaphores:
- Counting semaphore whose integer value can range over an unrestricted integer domain.
- Binary semaphore whose integer value cannot be > 1; can be simpler to implement.

Question # 10
The condition in which a set {P0, P1… Pn} of waiting processes must exist such that P0 is waiting for a resource that is held by P1, P1 is waiting for a resource that is held by P2, and so on, Pn-1 is waiting for a resource held by Pn, and Pn is waiting for a resource held by P0. This condition is known as _____.
Select correct option:
Mutual exclusion
Hold and wait
No preemption
Circular wait (Reference)

Question # 8
Deadlock detection and recovery technique is exactly similar to deadlock avoidance technique to handle deadlock in the system.
Select correct option:
-True
-False

Question # 9
In Overlay technique, we can overload any part of the program with the part of the program required needed recently.
Select correct option:
True
False

Question No: 8
Operating system is manages the use of hardware among the various application programs for the users.
True
False

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Question No: 9
In shortest job first algorithm CPU schedule process according to the,
Priority wise
CPU Burst
Distribute CPU resources equally among all processes
All of these

Question No: 5
In UNIX, a file descriptor is:
A positive integer associated with an open file. Its value describes the type of data in the file and the file location on disk.
A positive integer used to index the per process file descriptor table to eventually access an open file's attributes, including its location.
A positive integer used to index the per process file descriptor table, which points to the inode of the file containing the executable image of the process on disk.
All of the given choices.
None of the given choices.

A system call
Is an entry point into the kernel code
Allows a program to request a kernel service
Is a technique to protect I/O devices and other system resources
All of the these

Software may trigger an interrupt by executing a special operation called a system call (also called a monitor call).

Addresses generated *relative* to part of program, not to start of physical memory are
Virtual
Physical
Relocatable
Symbolic

The size of a page is defined by _____.
Select correct option:
CPU
Page Table
Physical Memory
Logical Memory

1. What scheduling policy will you use for each of the following cases? Explain your reasons for choosing them.
a. The processes arrive at large time intervals:
b. The system's efficiency is measured by the percentage of jobs completed.
c. All the processes take almost equal amounts of time to complete.

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Ans:
a. FCFS
b. STF
c. FCFS or RR

More MCQ's On Operating System

Multiple Choice Questions On Operating System Part-1

1) Routine is not loaded until it is called. All routines are kept on disk in a relocatable load format. The main program is loaded into memory & is executed. This type of loading is called _____
1 Static loading
2 Dynamic loading
3 Dynamic linking
4 Overlays
Ans) 3

2) Which of the following is crucial time while accessing data on the disk?
1 Seek time
2 Rotational time
3 Transmission time
4 Waiting time
Ans) 1

3) The host repeatedly checks if the controller is busy until it is not. It is in a loop that status register's busy bit becomes clear. This is called _____ and a mechanism for the hardware controller to notify the CPU that it is ready is called _____.
1 Interrupt and Polling
2 Polling and Spooling
3 Polling and Interrupt
4 Deadlock and Starvation
Ans) 3

4) Unix Operating System is an _____.
1 Time Sharing Operating System
2 Multi-User Operating System
3 Multi-tasking Operating System
4 All the Above
Ans) 4

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5) Which of the following memory allocation scheme suffers from External fragmentation?
1 Segmentation
2 Pure demand paging

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3 Swapping
4 Paging
Ans) 1

6) Information about a process is maintained in a _____.
1 Stack
2 Translation Lookaside Buffer
3 Process Control Block
4 Program Control Block
Ans) 3

7) Distributed OS works on the _____ principle.
1 File Foundation
2 Single system image
3 Multi system image
4 Networking image
Ans) 2

8) The problem of fragmentation arises in _____.
1 Static storage allocation
2 Stack allocation storage
3 Stack allocation with dynamic binding
4 Heap allocation
Ans) 4

9) Which file system does DOS typically use ?
1 FAT16
2 FAT32
3 NTFS
4 WNFS
Ans) 1

10) The program is known as _____ which interacts with the inner part of called kernel.
1 Compiler
2 Device Driver
3 Protocol
4 Shell
Ans) 4

11) The time taken by the disk arm to locate the specific address of a sector for getting information is called _____.
1 Rotational Latency
2 Seek Time
3 Search Time
4 Response Time
Ans) 2

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12) Which file system does Windows 95 typically use ?
1 FAT16
2 FAT32
3 NTFS
4 LMFS
Ans) 2

13) Identify the odd thing in the services of operating system.
1 Accounting
2 Protection
3 Error detection and correction
4 Dead lock handling
Ans) 3

14) Cryptography technique is used in _____.
1 Polling
2 Job Scheduling
3 Protection
4 File Management
Ans) 3

15) Which of the following is not advantage of multiprogramming?
1 Increased throughput
2 Shorter response time
3 Decreased operating system overhead
4 Ability to assign priorities to jobs
Ans) 3

16) In _____ OS, the response time is very critical.
1 Multitasking
2 Batch
3 Online
4 Real-time
Ans) 4

17) An optimal scheduling algorithm in terms of minimizing the average waiting time of a given set of processes is _____.
1 FCFS scheduling algorithm
2 Round robin scheduling algorithm
3 Shorest job - first scheduling algorithm
4 None of the above
Ans) 3

18) Real time systems are _____.
1 Primarily used on mainframe computers

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2 Used for monitoring events as they occur
3 Used for program development
4 Used for real time interactive users
Ans) 2

19) Which technique was introduced because a single job could not keep both the CPU and the I/O devices busy?
1 Time-sharing
2 SPOOLing
3 Preemptive scheduling
4 Multiprogramming
Ans) 4

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20) Inter process communication can be done through _____.
1 Mails
2 Messages
3 System calls
4 Traps
Ans) 2

21) In Priority Scheduling a priority number (integer) is associated with each process. The CPU is allocated to the process with the highest priority (smallest integer = highest priority). The problem of, Starvation ? low priority processes may never execute, is resolved by _____.
1 Terminating the process.
2 Aging
3 Mutual Exclusion
4 Semaphore
Ans) 2

22) CPU performance is measured through _____.
1 Throughput
2 MHz
3 Flaps
4 None of the above
Ans) 1

23) PCB =
1 Program Control Block
2 Process Control Block
3 Process Communication Block
4 None of the above
Ans) 2

24) Software is a program that directs the overall operation of the computer, facilitates its use and interacts with the user. What are the different types of this software ?
1 Operating system

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2 Language Compiler
3 Utilities
4 All of the above
Ans) 4

25) A _____ is a software that manages the time of a microprocessor to ensure that all time critical events are processed as efficiently as possible. This software allows the system activities to be divided into multiple independent elements called tasks.
1 Kernel
2 Shell
3 Processor
4 Device Driver
Ans) 1

26) The primary job of the operating system of a computer is to _____.
1 Command Resources
2 Manage Resources
3 Provide Utilities
4 Be user friendly
Ans) 2

27) With the round robin CPU scheduling in a time-shared system _____.
1 Using very large time slice degenerates in to first come first served algorithm
2 Using extremely small time slices improve performance
3 Using extremely small time slices degenerate in to last in first out algorithm
4 Using medium sized time slices leads to shortest request time first algorithm
Ans) 1

28) Which of the following is a criterion to evaluate a scheduling algorithm?
1 CPU Utilization: Keep CPU utilization as high as possible.
2 Throughput: number of processes completed per unit time.
3 Waiting Time: Amount of time spent ready to run but not running.
4 All of the above
Ans) 4

29) Which of the following is contained in Process Control Block (PCB)?
1 Process Number
2 List of Open files
3 Memory Limits
4 All of the Above
Ans) 4

30) Super computers typically employ _____.
1 Real time Operating system
2 Multiprocessors OS
3 desktop OS

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4 None of the above
Ans) 2

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31) Round robin scheduling is essentially the preemptive version of _____.
1 FIFO
2 Shortest job first
3 Shortes remaining
4 Longest time first
Right Ans) 1

32) A page fault occurs
1 when the page is not in the memory
2 when the page is in the memory
3 when the process enters the blocked state
4 when the process is in the ready state
Right Ans) 1

33) Which of the following will determine your choice of systems software for your computer ?
1 Is the applications software you want to use compatible with it ?
2 Is it expensive ?
3 Is it compatible with your hardware ?
4 Both 1 and 3
Right Ans) 4

34) Let S and Q be two semaphores initialized to 1, where P0 and P1 processes the following statements wait(S);wait(Q); ---; signal(S);signal(Q) and wait(Q); wait(S);---;signal(Q);signal(S); respectively. The above situation depicts a _____ .
1 Semaphore
2 Deadlock
3 Signal
4 Interrupt
Right Ans) 2

35) What is a shell ?
1 It is a hardware component
2 It is a command int:gasp:erpreter
3 It is a part in compiler
4 It is a tool in CPU scheduling
Right Ans) 2

36) Routine is not loaded until it is called. All routines are kept on disk in a relocatable load format. The main program is loaded into memory & is executed. This type of loading is called _____
1 Static loading
2 Dynamic loading
3 Dynamic linking
4 Overlays

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Right Ans) 3

37) In the blocked state
1 the processes waiting for I/O are found
2 the process which is runn:tongue:ing is found
3 the processes waiting for the processor are found
4 none of the above
Right Ans) 1

38) What is the memory from 1K - 640K called ?
1 Extended Memory
2 Normal Memory
3 Low Memory
4 Conventional Memory
Right Ans) 4

39) Virtual memory is _____.
1 An extremely large main memory
2 An extremely large secondary memory
3 An illusion of extremely large main memory
4 A type of memory used in super computers.
Right Ans) 3

40) The process related to process control, file management, device management, information about system and communication that is requested by any higher level language can be performed by _____.
1 Editors
2 Compilers
3 System Call
4 Caching
Right Ans) 3

41) If the Disk head is located initially at 32, find the number of disk moves required with FCFS if the disk queue of I/O blocks requests are 98,37,14,124,65,67.
1 310
2 324
3 315
4 321
Right Ans) 4

42) Multiprogramming systems _____.
1 Are easier to develop than single programming systems
2 Execute each job faster
3 Execute more jobs in the same time
4 Are used only on large main frame computers
Right Ans) 3

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43) Which is not the state of the process ?
1 Blocked
2 Running
3 Ready
4 Privileged
Right Ans) 4

44) The solution to Critical Section Problem is : Mutual Exclusion, Progress and Bounded Waiting.
1 The statement is false
2 The statement is true.
3 The statement is contradictory.
4 None of the above
Right Ans) 2

45) The problem of thrashing is effected scientifically by _____.
1 Program structure
2 Program size
3 Primary storage size
4 None of the above
Right Ans) 1

46) The state of a process after it encounters an I/O instruction is _____.
1 Ready
2 Blocked/Waiting
3 Idle
4 Running
Right Ans) 2

47) The number of processes completed per unit time is known as _____.
1 Output
2 Throughput
3 Efficiency
4 Capacity
Right Ans) 2

48) _____ is the situation in which a process is waiting on another process,which is also waiting on another process ... which is waiting on the first process. None of the processes involved in this circular wait are making progress.
1 Deadlock
2 Starvation
3 Dormant
4 None of the above
Right Ans) 1

49) Which of the following file name extension suggests that the file is Backup copy of another file ?

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1 TXT
2 COM
3 BAS
4 BAK
Right Ans) 4

50) Which technique was introduced because a single job could not keep both the CPU and the I/O devices busy?
1 Time-sharing
2 SPOOLing
3 Preemptive scheduling
4 Multiprogramming
Right Ans) 4

51) A critical region
1 is a piece of code which only one process executes at a time
2 is a region prone to deadlock
3 is a piece of code which only a finite number of processes execute
4 is found only in Windows NT operation system
Right Ans) 1

52) The mechanism that bring a page into memory only when it is needed is called _____
1 Segmentation
2 Fragmentation
3 Demand Paging
4 Page Replacement
Right Ans) 3

53) PCB =
1 Program Control Block
2 Process Control Block
3 Process Communication Block
4 None of the above
Right Ans) 2

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54) FIFO scheduling is _____.
1 Preemptive Scheduling
2 Non Preemptive Scheduling
3 Deadline Scheduling
4 Fair share scheduling
Right Ans) 2

55) Switching the CPU to another Process requires to save state of the old process and loading new process state is called as _____.
1 Process Blocking
2 Context Switch

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3 Time Sharing
4 None of the above
Right Ans) 2

56) Which directory implementation is used in most Operating System?
1 Single level directory structure
2 Two level directory structure
3 Tree directory structure
4 Acyclic directory structure
Right Ans) 3

57) The Banker¿s algorithm is used
1 to prevent deadlock in operating systems
2 to detect deadlock in operating systems
3 to rectify a deadlocked state
4 none of the above
Right Ans) 1

58) A thread
1 is a lightweight process where the context switching is low
2 is a lightweight process where the context swithching is high
3 is used to speed up paging
4 none of the above
Right Ans) 1

59) _____ is a high level abstraction over Semaphore.
1 Shared memory
2 Message passing
3 Monitor
4 Mutual exclusion
Right Ans) 3

60) A tree sturctured file directory system
1 allows easy storage and retrieval of file names
2 is a much debated unecessary feature
3 is not essential when we have millions of files
4 none of the above
Right Ans) 1

---

1) A 20-bit **address bus** allows access to a memory of capacity
(a) 1 Mb (b) 2 Mb (c)32Mb (d) 64 Mb
Answer:- a).

2) On-chip cache has

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(a) Lower access time than RAM (b) larger capacity than off chip cache
(c) Its own data bus (d) become obsolete
Answer:- a).

3) A RAID system is useful because
(a) It increases processor speed (b) increases disk storage capacity
(c) Increases disk storage capacity and availability (d) increases OS efficiency
Answer:- c).

4) Multiprogramming refers to
(a) Having several programs in RAM at the same time (b) multitasking
(c) Writing programs in multiple languages (d) none of the previous
Answer:- a).

5) Multiprocessing is
(a) Same as Multitasking (b) same as multiprogramming
(c) Multiuser (d) involves using more than one processor at the same time
Answer:- d).

6) Timesharing is the same as
(a) Multitasking (b) multiprogramming
(c) Multiuser (d)none of the previous
Amswer:- a).

7) The average memory access time for a machine with a cache hit rate of 90% where the cache access time is 10ns and the memory access time is 100ns is
(a) 55ns (b) 45ns (c) 90ns (d) 19ns
Answer:- d).

8) The memory address register is used to store
(a) Data to be transferred to memory
(b) Data that has been transferred from memory
(c) The address of a memory location
(d) An instruction that has been transferred from memory.
Answer:- c).

9) The memory data register is used to store
(a) Data to be transferred to or from memory
(b) Data to be transferred to the stack
(c) The address of a memory location
(d) An instruction that has been transferred from memory
Answer:- a).

10) In accessing a disk block the longest delay is due to

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(a) Rotation time (b) seek time (c) transfer time (d) clock speed.
Answer :- b).

11) Consider the following set of processes, with the length of the CPU-burst time given in milliseconds:

Process	Burst Time	Priority
P1	10	3
P2	1	1
P3	2	3
P4	1	4
P5	5	2

Assume that P1 is at the head of the ready queue and P5 is at the tail, and ignore the time for changing inter-processes. Using Round Robin scheduling algorithm (where the quantum q= 1ms) which is the average turnaround time?

a) tav = 12ms b) tav = 9.2ms c) tav = 7ms d) tav = 10.8ms e) None of these
Answer:- b)

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12) There are 10 different processes running on a workstation. Idle processes are waiting for an input event in the input queue. Busy processes are scheduled with the Round-Robin timesharing method. Which out of the following quantum times is the best value for small response times, if the processes have a short runtime, e.g. less than 10ms?

A) tQ = 15ms **B)** tQ = 40ms **C)** tQ = 45ms **D)** tQ = 50ms
Answer:- a).

13) Consider a swapping system in which memory consists of the following hole sizes in memory order:

H0	H1	H2	H3	H4	H5	H6	H7
10K	4KB	20KB	18KB	7KB	9KB	12KB	15KB

and a successive segment request of
a) 12 KB
b) 10KB
c) 9KB
Which of the following sentences is/are true?
a) First Fit algorithm allocates H2, H0, H3 for the mentioned request.
b) Best Fit algorithm allocates H2, H0, H3 for the mentioned request.
c) First Fit algorithm allocates H2, H6, H7 for the mentioned request.
d) Worst Fit algorithm allocates H2, H3, H6 for the mentioned request.
Answer:- a).

14) Page fault occurs when
a) The page is corrupted by application software.
b) The page is in memory.
c) The page is not in memory
d) One tries to divide a number by 0.
Answer:- c)

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15) Determine the number of page faults when references to page occur in the following order: 1,2,4,5,2,1,2,4. assume that the main memory can accommodate 3 pages and the main memory already has the pages 1 and 2, with page 1 having been brought earlier than page 2. (LRU algorithm is used)
a) 3 b) 5 c) 4 d) none of these.
Answer:- c)

16) The page replacement policy that sometimes leads to more page faults when the size of the memory is increased is
a) FIFO b) LRU c) no such policy exists d) none of the above.
Answer:- a).

17) DIJKSTRA'S baking algorithm in an operating system, solves the problem of
a) Deadlock avoidance b) deadlock recovery
c) Mutual exclusion d) context switching.
Answer:- a)

18) Necessary conditions for deadlock are
a) Non pre-emption and circular waitb) mutual exclusion ad partial allocation
c) Both a and b d) none of the above.
Answer:- c).

19) At a particular time, the value of counting semaphore is 10. it will become 7 after
a) 3 V operations b) 5P operations c) 5V and 2P operations
d) 13 P and 10 V operations.
Answer:- d).

20) Semaphores are used to solve the problem of
a) Race condition b) process synchronization
c) Both of above d) none of the above.
Answer: – b).

21) The size of virtual memory depends on
a) The size of the data bus b) the size of main memory
c) The size of address bus d) none of the above.
Answer:- c).

22) Suppose that a process is in 'BOCKED' state waiting for some I/O service. When the service is completed, it goes to the
a) RUNNING state b) READY state c) SUSPENDED state d) TERMINATED state.
Answer:- b)

23) In real time operating systems, which of the following is the most suitable scheduling scheme?
a) round- robin b) FCFSc) pre-emptive scheduling d) random scheduling
answer:- c).

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24) Which of the following is well suited for batch processing.
a) Process control b) video game control
c) Preparing pay bills of employees d) none of the above.
Answer:- c).

25) Which of the following page replacement algorithms suffers from belady's anomaly?
a) Optimal replacement b) LRU
c) FIFO d) both optimal replacement and FIFO.
Answer:- c).

True or False, circle T or F.

T a. A binary semaphore takes on numerical values 0 and 1 only.
T b. An atomic operation is a machine instruction or a sequence of instructions that must be executed to completion without interruption.
F c. Deadlock is a situation in which two or more processes (or threads) are waiting for an event that will occur in the future. (cannot occur)
T d. Starvation is a situation in which a process is denied access to a resource because of the competitive activity of other, possibly unrelated, processes. (denied access indefinitely, possibly infinitely)

F e. While a process is blocked on a semaphore's queue, it is engaged in busy waiting.
T f. Circular waiting is a necessary condition for deadlock, but not a sufficient condition.
(a condition for the deadlock to occur)
F g. Mutual exclusion can be enforced with a general semaphore whose initial value is greater than 1.
(equal to 1)
T F h. External fragmentation can occur in a paged virtual memory system.
(question was too vague, either answer was accepted)
T i. External fragmentation can be prevented (almost completely) by frequent use of compaction, but the cost would be too high for most systems.
T j. A page frame is a portion of main memory.
F k. Once a virtual memory page is locked into main memory, it cannot be written to the disk.
(A locked page cannot be swapped out, but "swapped out" is more than "written to".)
F l. Pages that are shared between two or more processes can never be swapped out to the disk.
(sharing does not require locking)
F m. The allocated portions of memory using a buddy system are all the same size.
F n. Demand paging requires the programmer to take specific action to force the operating system to load a particular virtual memory page.
T o. Prepaging is one possibility for the fetch policy in a virtual memory system.
T p. The resident set of a process can be changed in response to actions by other processes.
F q. The working set of a process can be changed in response to actions by other processes.

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F r. The translation look aside buffer is a software data structure that supports the virtual memory address translation operation. (hardware)

F s. In a symmetric multiprocessor, threads can always be run on any processor.
(from the hardware's point of view only, this would be true, but the OS and user can put restrictions on processor assignment.)

F t. Thrashing will never be a problem if the system has 1 GB of real memory.

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In Case of mistakes in file please let me know